

For the Use of a Registered Medical Practitioner or a Hospital or Laboratory only.



MELONEX

(Meloxicam Tablets 7.5mg and 15mg)

Meloxicam is an oxicam derivative and belongs to enolic acid group of nonsteroidal anti-inflammatory drugs (NSAIDs).

Description

Melonex-7.5 : Yellow coloured, round, flat, uncoated tablet embossed with 'INTAS' on one side and bisecting line on other side of the tablet.

Melonex-15 : Yellow coloured, round, flat, uncoated tablet embossed with 'INTAS' on one side & bisecting line on other side of the tablet.

Composition

- Melonex-7.5
Each tablet contains :
Meloxicam BP.....7.5 mg
- Melonex-15
Each tablet contains :
Meloxicam BP.....15 mg

Clinical Pharmacology

Mechanism of Action

Meloxicam is a nonsteroidal anti-inflammatory drug (NSAID) that exhibits anti-inflammatory, analgesic, and antipyretic properties in animals. Meloxicam showed potent antiinflammatory activity in all standard models of inflammation. NSAIDs are believed to exert their pharmacologic effects primarily through inhibition of the enzyme of the cyclooxygenase (COX). In turn, inhibition of this enzyme leads to an inhibition of biosynthesis of prostaglandins and other autacoids, substances which are potent biological mediators involved in diverse physiologic functions as well as pathologic conditions. Meloxicam has been shown to inhibit COX-2 in several in vitro and ex vivo test systems. The inhibition of thromboxane in platelets, and consequently platelet aggregation, occurs via inhibition of COX-1. Meloxicam inhibition of thormboxane in platelets (via COX-1) is dose dependent and incomplete at anti-inflammatory doses. No significant inhibition of platelet aggregation has been observed with meloxicam at the recommended therapeutic doses of 7.5 and 15 mg once daily. Inhibition of COX-2 also inhibits the production of systemic prostacyclin. Inhibition of prostacyclin may have a prothrombotic effect.

Pharmacokinetics

Absorption

The absolute bioavailability of meloxicam capsules was 89% following a single oral dose of 30 mg compared with 30 mg IV bolus injection. Meloxicam capsules have been shown to be bioequivalent to meloxicam tablets. After multiple oral doses the pharmacokinetics of meloxicam capsules were dose proportional over the range of 7.5 mg to 15 mg. Mean Cmax was achieved within four to five hours after a 7.5 mg meloxicam tablet was taken under fasted conditions, indicating a prolonged drug absorption. The rate or extent of absorption was not affected by multiple dose administration, suggesting linear pharmacokinetics. With multiple dosing, steady state conditions were reached by day 5.

Distribution

The mean volume of distribution (Vss) of meloxicam is approximately 10 L. Meloxicam is approximately 99% bound to human plasma proteins (primarily albumin). Meloxicam penetration into human red blood cells, after oral dosing, is less than 10%. Following a single oral dose, meloxicam concentrations in synovial fluid range from 40% to 50% of those in plasma. The free fraction in synovial fluid is 2.5 times higher than in plasma due to the lower albumin content in synovial fluid as compared to plasma. The significance of this penetration is not known. The volume of distribution is increased in severe renal failure.

Metabolism

Meloxicam is almost completely metabolized mainly by oxidation to four pharmacologically inactive metabolites. The major metabolite is 5'-carboxy meloxicam, which is formed by oxidation of an intermediate metabolite 5'-hydroxymethyl meloxicam, which is also excreted to a lesser extent. In vitro studies indicate that cytochrome P-450 2C9 plays an important role in this metabolic pathway with a minor contribution of the CYP 3A4 isozyme. The peroxidase activity is believed to be responsible for the formation of other two metabolites.

Excretion

Meloxicam is excreted predominantly in the form of metabolites. It is excreted in similar amounts in the urine and in the feces. Only traces of the unchanged parent compound are excreted in the urine and feces. There is significant biliary and/or enteral secretion of the drug. The mean plasma elimination half-life ($t_{1/2}$) ranges from 15 hours to 20 hours.

Special Populations

Pediatric

The pharmacokinetics of meloxicam in pediatric patients under 18 years of age have not been investigated.

Hepatic Insufficiency

No dose adjustment is necessary in patients with mild to moderate hepatic insufficiency. There is no adequate data regarding the use of meloxicam in patients with severe hepatic impairment (Child-Pugh Class III).

Renal Insufficiency

No dose adjustment is necessary in patients with mild to moderate renal failure (creatinine clearance >15 mL/min). The use of meloxicam in subjects with severe renal impairment is not recommended.

Hemodialysis

Hemodialysis does not lower the total drug concentration in plasma; therefore, additional doses are not necessary after hemodialysis. Meloxicam is not dialyzable.

Indications and Usage

Meloxicam is indicated for relief of the signs and symptoms of

- Osteoarthritis
- Rheumatoid arthritis
- Ankylosing spondylitis

Dosage and administration

Meloxicam may be taken without regard to timing of meals.

Osteoarthritis

7.5mg per day. If necessary the dose may be increased to 10 mg/day.

Rheumatoid arthritis

15 mg/day. According to therapeutic response, the dose may be reduced to 7.5 mg/day.

Ankylosing spondylitis

In patients with increased risks of adverse reactions: Start treatment at the dose of 7.5 mg/day.

In dialysis patients with severe renal failure : The dose should not exceed 7.5 mg/day. The total daily dosage of meloxicam should not exceed 15 mg.

As a dosage for used in children has yet to be established, usage should be restricted to adults.

Tablets should be swallowed with water or other fluid in conjunction with food.

Contraindications

Meloxicam is contraindicated in patients-

- With known hypersensitivity to meloxicam or any excipient of the product. There is a potential for cross sensitivity to acetylsalicylic acid and other non-steroidal anti-inflammatory drugs (NSAIDs).
- Meloxicam should not be given to patients who have developed signs of asthma, nasal polyps, angio-oedema or urticaria following the administration of acetyl-salicylic or other NSAIDs.
- Active peptic ulceration.
- Severe hepatic insufficiency.
- Non-dialysed severe renal insufficiency.
- Children and adolescents aged less than 15 years.
- Pregnancy or breast feeding.

Precautions and Warnings

As with other NSAIDs caution should be exercised when treating patients with a history of gastro-intestinal disease and in patients receiving treatment with anticoagulants. Patients with gastro-intestinal symptoms should be monitored. Meloxicam should be withdrawn if peptic ulceration or gastro-intestinal bleeding occurs.

Gastro-intestinal bleeding, ulceration or perforation can occur at any time during treatment, with or without warning symptoms or a previous history of serious gastro-intestinal events. The consequences of such events are generally more serious in the elderly.

Special attention should be paid in patients reporting mucocutaneous adverse events and consideration given to discontinuing meloxicam.

NSAIDs inhibit the synthesis of renal prostaglandins which play a supportive role in the maintenance of renal perfusion. In patients whose renal blood flow and blood volume are decreased, administration of an NSAID may precipitate overt renal decomposition which is typically followed by recovery to pretreatment state upon discontinuation of non-steroidal anti-inflammatory therapy. Patients at greatest risk of such a reaction are dehydrated patients, those with congestive heart failure, liver cirrhosis, nephritic syndrome and overt renal disease, those receiving a diuretic or those having undergone major surgical procedures, which led to hypovolemia. In such patients the volume of diuresis and the renal function should be carefully monitored at the beginning of therapy.

In rare instances NSAIDs may be the cause of interstitial nephritis, glomerulo-nephritis, renal medullary necrosis or nephrotic syndrome.

The dose of meloxicam in patients with end-stage renal failure on haemodialysis should not be higher than 7.5 mg. No dose reduction is required in patients with mild or moderate renal impairment (i.e in patients with a creatinine clearance of greater than 25 mL/min).

As with most other NSAIDs, occasional elevation of serum transaminases or other parameters of liver function have been reported. In most cases these have been small and transient increases above the normal range. If the abnormality is significant or persistent, meloxicam should be stopped and follow up tests carried out.

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No dose reduction is required in patients with clinically stable liver cirrhosis. Frail or debilitated patients may tolerate side effects less well and such patients should be carefully supervised. As with other NSAIDs, caution should be used in the treatment of elderly patients who are more likely to be suffering from impaired renal, hepatic or cardiac function. Induction of sodium, potassium and water retention and interference with the natriuretic effects of diuretics may occur with NSAIDs. Cardiac failure or hypertension may be precipitated or exacerbated in susceptible patients as a result. There are no specific studies about effects on the ability to drive vehicles and to use machinery. Patients who experience visual disturbances, drowsiness or other central nervous system disturbances should refrain from these activities.

WARNINGS

RISK OF GI ULCERATION, BLEEDING AND PERFORATION WITH NSAID.

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (eg. Dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious GI events and other risk factors associated with peptic ulcer disease (eg. Alcoholism, smoking and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Drug Interactions

Other NSAIDs including salicylates in high doses; Concomitant administration of more than one NSAID may increase the risk of gastro-intestinal ulceration and bleeding through synergistic action.

Oral anticoagulant, ticlopidine, systemically administered heparin, thrombolytics, increased risk of bleeding. If such co-prescribing cannot be avoided, close monitoring of the effects of anticoagulants is required.

Lithium; NSAIDs have been reported to increase lithium plasma levels. It is recommended that plasma lithium levels be monitored when initiating, adjusting and discontinuing meloxicam.

Methotrexate; As other NSAIDs meloxicam may increase the hematologic toxicity of methotrexate. In this situation, strict monitoring of blood cell count is recommended.

Contraception; NSAIDs have been reported to decrease the efficacy of intrauterine devices.

Diuretics; Treatment with NSAIDs is associated with the potential for acute renal insufficiency in patients who are dehydrated. Patient receiving meloxicam and diuretics should be adequately hydrated and be monitored for renal function prior to initiating treatment.

Antihypertensives : (eg. beta-blockers, ACE-inhibitors, vasodilators, diuretics); A reduced effect of the antihypertensive drug by inhibition of vasodilating prostaglandins has been reported during treatment with NSAIDs.

Cholestyramine binds meloxicam in the gastro-intestinal tract leading to a faster elimination of meloxicam.

Nephrotoxicity of cyclosporin may be enhance by NSAIDs via renal prostaglandin mediated effects. During combined treatment renal function is to be measured.

No relevant pharmacokinetic drug-drug interactions were detected with respect to the concomitant administration of antacids, cimetidine, digoxin and furosemide. Interactions with oral anti diabetics can not be excluded.

Pregnancy

There are no adequate and well controlled studies in pregnant women. Meloxicam should be used during pregnancy only if potential benefit outweighs the potential risk to the fetus.

Nursing mothers

It is not known whether meloxicam is excreted in human milk. So considering the potential for serious adverse reactions in nursing infants, meloxicam should be administered by taking into account the importance of the drug to the mother.

Pediatric use

Safety and efficacy of meloxicam have not been established in patients below the age of 18 years.

Geriatric use

As with any NSAID, caution should be exercised in treating the elderly (65 years and older).

Adverse effects

Meloxicam is generally well tolerated. The adverse reactions that may appear are:

Gastro-intestinal : Abdominal pain, constipation, diarrhea, dyspepsia, flatulence, nausea, oesophagitis, stomatitis and vomiting. Rarely, peptic ulceration, perforation or gastro-intestinal bleeding, sometimes severe, particularly in the elderly may occur. Rare cases of gastritis have been reported.

Haematological : Reports of blood count disturbances, anaemia, leucocytopenia and thrombocytopenia. There are isolated reports on agranulocytosis in patients taking meloxicam and other drugs with myelotoxic potential.

Dermatological : Skin rash, pruritus, urticaria and photosensitivity reactions. Very rarely, erythema multiform, Stevens-Johnsons syndrome and toxic epidermal necrolysis or other bullous reactions.

Respiratory : Patients allergic to aspirin or other NSAIDs have experienced asthma attacks.

Central Nervous System : Drowsiness, headache, lightheadedness, tinnitus and vertigo.

Cardiovascular : Oedema (including that of the lower limbs), elevated blood pressure, palpitations and flushes.

Genito-urinary : Laboratory tests of renal function may be disturbed, such as elevated creatinine or urea.

Vision Disorders : Rare reports on visual disturbances, including blurred vision.

Hypersensitivity Reactions : Rare reports on angioedema and anaphylactic or anaphylactoid reactions.

Hepatic system : Liver function test may show transitory disturbance such as elevation of bilirubin or transaminase levels.

Effects on the ability to drive and use machines.

There are no specific studies about effects on the ability to drive vehicles and to use machinery. Patients who experience visual disturbances, drowsiness or other central nervous system disturbances should refrain from these activities.

Overdosage

In case of overdosage, patients should be managed with symptomatic and supportive care. In cases of acute overdose, gastric lavage followed by activated charcoal is recommended. For substantial overdose or severely symptomatic patients, activated charcoal may be administered repeatedly. Administration of cholestyramine may be useful following an overdose.

Storage Condition:

Store below 30°C in a dark dry place

Shelf Life

3 years from the date of manufacturing

Registration holder

JETPHARMA SDN BHD

Presentation

Melonex-7.5 & Melonex-15 are available in Alu/Alu blister strips of 10 tablets, & 10 such strips are packed in a printed carton. (10 x 10T).

Manufactured by

INTAS

INTAS PHARMACEUTICALS LTD.

Matoda - 382 210, Dist. Ahmedabad, INDIA.

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